

Artificial Intelligence based Intelligent Stethoscope: A Low-Cost Device based on Body Auscultation for Early Medical Diagnosis of Heart, Lung and Prenatal Health



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Cardiovascular and pulmonary diseases are critical diseases and need to be diagnosed as early as possible. There is a lack of medical professionals in remote areas to diagnose these diseases. An Artificial Intelligence based automatic intelligent stethoscope is developed to diagnose cardiac, pulmonary and prenatal disorders to minimize human intervention. The proposed system involves a convolutional neural network (1D-CNN) model to diagnose the cardiac disorders from the PCG signal and lungs disorder from pulmonary signals because of its high accuracy and robustness. To improve the accuracy in a noisy environment and make the method robust, the proposed method has used data augmentation techniques for training and multi-classification of cardiac diseases. The proposed model is ported into small computing devices to make a stand-alone diagnostic tool that may be of help in remote primary health care centers. The developed prototype contains a stethoscope, sound card, display screen, DSP processor, and power bank. A user friendly GUI application is developed which is based on python for viewing and analysis of generated reports. The proposed method is non-invasive, efficient, robust, and has low time complexity making it suitable for real-time applications.

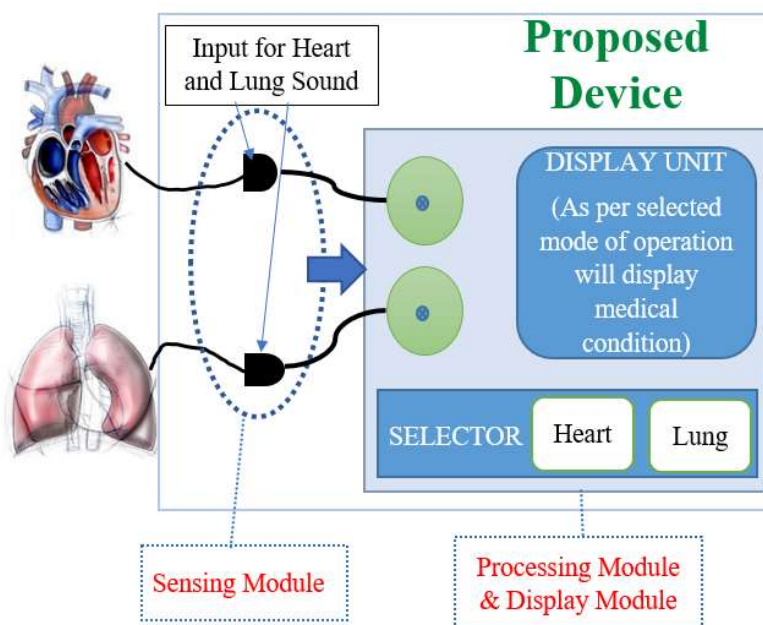


Fig. 1: Flowchart of the proposed method.

This device uses the technology based on artificial intelligence where machine learning and deep learning is used for training the model for heart, lungs and prenatal sound. The developed device integrates a low-cost DSP processor with a screen and stethoscope. Its aim is to diagnose heart, lungs and prenatal health and classify the patient's (heart, lungs and prenatal sound) as normal or abnormal with type of disease.

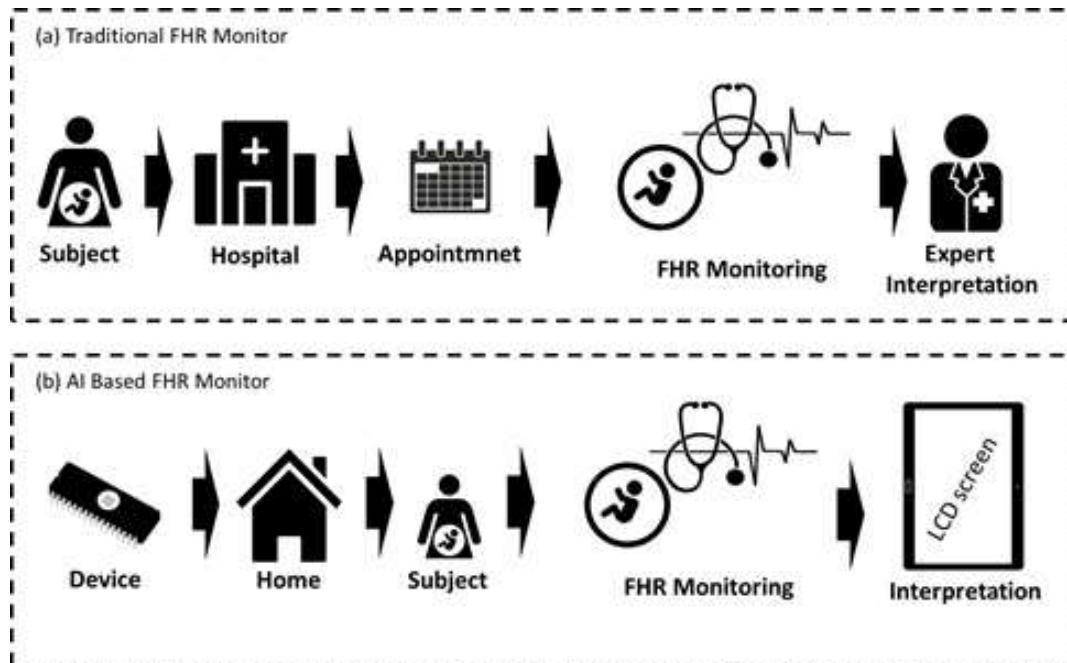


Fig. 2: Fetal Heart rate monitoring during pregnancy

Fetal heart rate (FHR) is used to monitor the fetal state by obstetricians as a screening tool. Common guidelines for visual interpretation of FHR signals results in significant subjective variability due to the fetal physiological dynamics complexity. Automated diagnostic technology can assist obstetricians in medical decisions based on artificial intelligence and also can be an automatic diagnostic tool for primary health care centres and remote areas.

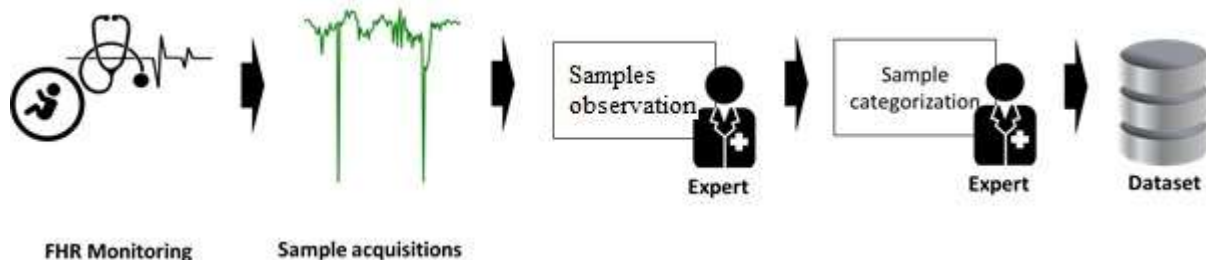


Fig. 3: CTU-UHB dataset preparation

It is a low-cost, real time and small portable intelligent device. These type of devices are not available in the market. This device is absolutely new where automatic diagnosis of heart, lungs, prenatal health can be performed without trained manpower. This device will have a great impact in the society due to its small size and low budget. It is based on artificial intelligence where machine learning and deep learning is used for training the model for heart, lungs and prenatal sound. It can be used for the screening purpose in the remote locations, primary health care centers, health care camps, medical clinics and personal use.

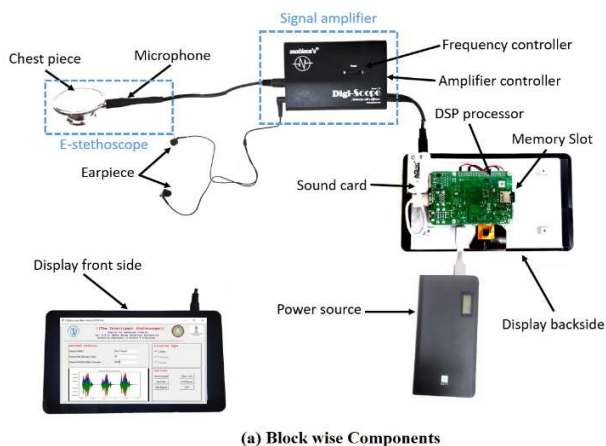


Fig 4: Developed prototype presenting its components: a low-cost DSP processor with a touch-screen, rechargeable battery, sound card and a stethoscope

Specifications:

- Device dim approximately 19cm*11cm*3cm
- Ultra low power ~ 1W
- 16bit per sample,
- 8000 Hz sampling rate
- 128kbps Bitrate
- Gaussian butter worth filter
- Proper GUI user interface
- 10 Finger Capacitive Touch
- Diagnose heart, lungs and health

Working Process:

- Fill patients details (Name, Age, Gender)
- Place stethoscope.
- Press record button on GUI
- Record and processing time: 5-10 sec
- View graph
- Press generate report
- View detailed report

The device also has a Graphical User Interface (GUI) facility on which details of the patient can be entered. It generates a report stating the patient's heart condition. Prototype is practically usable and affordable, making it well suited for medical screening at primary health care centers and remote areas in India.

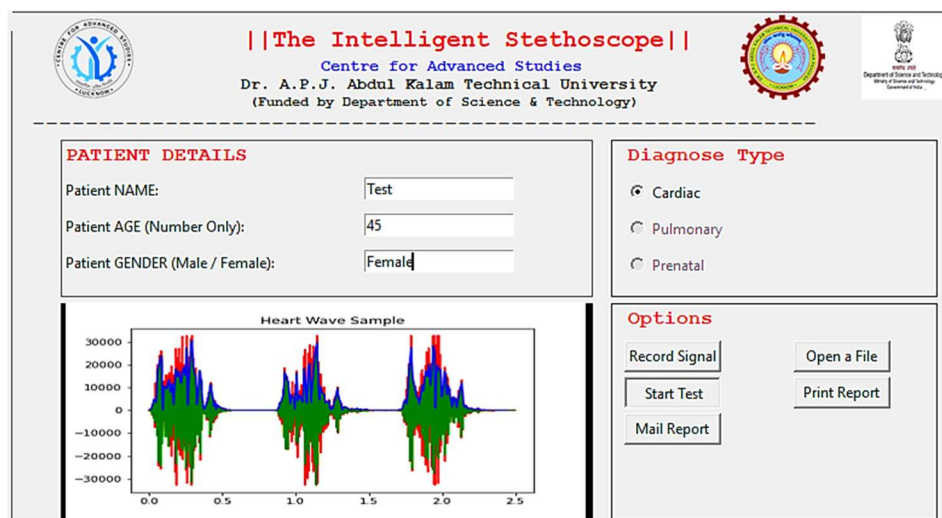


Fig 5: The intelligent stethoscope user interface



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“It is a low-cost electronic hardware integrated with the proposed methodology which can perform in real-time and can achieve high accuracy and reliability. It can be used to support the expert decision and as an automatic stand-alone diagnostic tool which can be a potential breakthrough within the Biomedical Engineering in India.” – Prof. M.K. Dutta (Principal Investigator)



• Research Team:



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• Patents from the project:

- (A) “A System and Method for Machine Learning Based Automatic Detection of Abnormality from Cardiac Sound” Application number: 201911033584 Dated. 21.08.2019, Status: Applied and under Examination
- (B) “Artificial Intelligence based System and Method for Multi-Class Classification of Heart Diseases”, Allotted Application number 201911047767 Dated: 22.11.2019, Status: Applied and under Examination
- (C) “Automatic Detection of Lungs Disease using Machine Learning from Respiratory Sound” Application number: 202011009043 Dated. 03.03.2020, Status: Applied and under Examination

• Publications from the project:

- 1) Anjali Yadav, et al. "Machine learning-based classification of cardiac diseases from PCG recorded heart sounds." Neural Computing and Applications (2019): 1-14.
- 2) Neeraj Baghel, Malay Kishore Dutta, Radim Burget, “Automatic Diagnosis of Multi-Cardiovascular Diseases from PCG Signals using Deep Convolutional Neural Network”, Computer Methods and Programs in Biomedicine, Elsevier DOI: <https://doi.org/10.1016/j.cmpb.2020.105750> [2020].
- 3) Neeraj Baghel, Vivek Nangia, Malay Kishore Dutta, “ALSDNET: Automatic Lungs Sound Diagnosis Network from Pulmonary Signals” Neural Computing and Applications, under review Springer
- 4) Neeraj Baghel, Malay Kishore Dutta, Radim Burget, “FHR-Net: Automatic Diagnosis of Fetal Acidosis from Fetal Heart Rate Signals using 1D-CNN”, Biomedical Signal Processing and Control, under revision.
- 5) Juwairiya Siraj Khan, Manoj Kaushik, Vinay Kumar Pathak, Malay Kishore Dutta, “PCGNet: Deep learning and Power Spectrogram based automatic diagnosis of multiple cardiac diseases using phonocardiogram signals”, Biomedical Signal Processing and Control, under revision.